

Resonance-Block Summation of Transseries

Small divisors, coalescing Stokes actions,
and nonlinear inversion

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Abstract

A countable Stokes expansion can have locally finite, finitely generated positive action support and still fail to converge at every positive value of the large variable. We establish this obstruction, and a constructive repair, for the meromorphic Borel kernels

$$\widehat{F}_{\omega}(t) = \frac{t^d}{\prod_{j=1}^d \sin(\pi\omega_j t)}, \quad \omega_j > 0.$$

Nearby poles are grouped into blocks of at most d nodes. Each block is a confluent divided difference and admits a bound independent of every internal pole separation. The block series equals the lateral Stokes jump, converges normally throughout the right half-plane, and has an explicit uniform action-cutoff error. For two incommensurate lattices, the ungrouped series has the exact abscissa determined by exponential Diophantine approximation; this abscissa can be infinite even though the original series has exact Gevrey order one. We also prove a uniform coalescence profile, characterize rational-exponential compression, and transport the block sums through analytic inversion with an explicit geometric tail. These results supply a concrete extension beyond a weighted individual-sector hypothesis in the ProveIt repository. Complete conventional proofs and reproducible checks are included; the claims are not Lean-formalized and independent research priority is not asserted.

Repository snapshot: 0c973d8f5e7ef4550f4df1bf486d890037fd9f14.

Source, PDF, verification programs, executed results, and build instructions are distributed together. No repository files were modified.

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1 The question and the contribution

Editorial note (ProveIt, 2026-09-29). Scope. The title is broader than the results. Every theorem below concerns the sine-product Borel kernels (2.1), with a fixed finite number d of period lattices; the one statement beyond them, the bounded-cluster criterion of Proposition 9.1, holds only under its stated hypotheses. No general resonance-block theory for transseries is proved. The “resonances” here are near-coincident Borel poles, not the equal-action resonances of the Stokes-transport and moving-fold packages of this series.

1.1 A gap between formal support and analytic realization

A familiar transseries calculation assigns an exponentially small term to each Borel singularity. When the singularities lie at positive actions a , the expected Stokes correction has the schematic form

$$J(x) = 2\pi i \sum_a p_a(x) e^{-ax}, \quad x \rightarrow +\infty. \quad (1.1)$$

The polynomial p_a records the principal part of a pole. Three different questions must be kept separate. Is the support legitimate for formal asymptotic bookkeeping? Does a function with these asymptotics exist? Does the displayed infinite sum converge numerically at fixed x ? In the model studied here the first two answers are yes, while the third can be no for every positive x .

The obstruction does not involve actions accumulating at a finite point. Our actions form a finite union of positive lattices. Every bounded action interval contains finitely many poles, and all actions belong to the additive monoid generated by $1/\omega_1, \dots, 1/\omega_d$. The obstruction is instead that distinct lattices can have extremely close nodes at arbitrarily large actions. Their individual residues contain small denominators. The corresponding finite group of residues cancels these denominators, but an infinite expansion that estimates the residues separately does not see the cancellation.

This observation leads to a concrete extension problem:

Can one construct a normally convergent Stokes representation, with explicit uniform error bounds and a usable nonlinear inversion calculus, without imposing a Diophantine condition on the periods?

The answer is affirmative for every fixed finite number of lattices in (2.1) below. We also determine exactly when the ungrouped representation converges in the two-lattice case.

1.2 Relationship to the repository

The repository’s article *Nonlinear Stokes Transport under Logarithmic Inversion* proves a contour-based inverse formula and a countable extension under a weighted summability assumption [1]. Its label `cor:countable` requires the sum of individual perturbation norms to be finite and sufficiently small. Its finite-action collision proposition, `prop:collision`, treats bounded amplitudes that combine by addition when actions coincide. These are appropriate hypotheses and conclusions; this report does not correct an error in them.

The present problem starts one step earlier. We obtain perturbations from Borel residues whose amplitudes can individually diverge as two poles approach one another. The weighted individual-residue norm can be infinite. A finite resonance block, unlike either of its constituent residues, has a controlled analytic norm. We prove the estimate that makes the repository’s analytic inversion method applicable after this change of representation. The inverse coefficient formula itself remains classical.

Editorial note (ProveIt, 2026-09-29). This report answers, without citing it, the forward half of the Stokes-transport article’s research question “Countably many resurgent

input poles” (Nonlinear_Stokes_Transport_Logarithmic_Inversion/article.tex in Analysis/Transseries/docs/series-and-transseries/, the question following `cor:countable` in its list of research questions; this report inspected only lines 180–360 of that source). That question proposes normal convergence of the forward meromorphic Borel kernel as a starting hypothesis. For the kernels (2.1), Corollary 4.3 shows that normal convergence of the individual residues can fail at every positive argument, and Theorem 3.4 restores it for resonance blocks with uniform constants. The resurgent half, an exact-core resurgent correction for the countable inverse, is not answered here (project 6 of Section 11). Theorem 8.1 is that article’s `thm:transport` in core coordinates, as stated in Section 8.

The series-and-transseries inventory also contains recent studies of finite-action folds, arithmetic action accumulation, and inverse harmonic Stokes data [2]. The mechanism here is distinct: the number of lattices is finite, their positive poles have no finite accumulation, and the near-collisions occur at unbounded action. The source comparison is limited to the inspected inventory and the cited Stokes article. It is not a claim-by-claim audit of the repository’s entire corpus.

1.3 Classical ingredients and the boundary of the novelty claim

Borel–Laplace summation and nonlinear closure are established subjects; Sauzin’s exposition is a suitable reference [3]. Divided differences, including confluent nodes and their integral representations, are classical [4]. Small-divisor phenomena for quantum logarithms and parameter-dependent functional equations are treated by Marmi and Sauzin [5]. Meromorphic Borel transforms, residue expansions, and Stokes jumps in quantum-dilogarithm problems are treated explicitly by Garoufalidis and Kashaev [6].

Accordingly, neither cancellation of coalescing poles nor the general existence of useful integral representations when some separated expansions fail is claimed as a discovery. The contribution developed here is a specific quantitative package: finite block cardinality, explicit separation-independent bounds in arbitrary finite dimension, the exact raw-series threshold for two lattices, a uniform collision law, rational-exponential rigidity, and a blockwise inverse certificate. The proofs establish the stated results. Whether every refinement is new to the literature requires a broader priority review.

1.4 Results at a glance

The central theorem is Theorem 3.4: for a block C with smallest action a_C and $m_C \leq d$ pole occurrences, its contribution satisfies

$$|S_C(z)| \leq K(1 + a_C)^d e^{-a_C \operatorname{Re} z} \sum_{j=0}^{m_C-1} \frac{|z|^j}{j! \rho^{m_C-1-j}}.$$

All constants are explicit and uniform on a compact positive period box. No factor involves the separation of two nodes in the same block. Theorem 4.2 identifies the ungrouped abscissa as

$$\beta(\alpha) = \limsup_{n \rightarrow \infty} \frac{1}{n} \log \frac{1}{\operatorname{dist}(n\alpha, \mathbb{Z})}.$$

Theorems 6.1, 7.2, and 8.1 then give, respectively, the collision profile, the exact compression criterion, and nonlinear transport. Section 11 records further questions rather than treating these finite-lattice results as a complete theory of countable-action resurgence.

2 The Borel model and its lateral sums

2.1 Conventions

Fix $d \geq 1$ and positive real periods $\omega = (\omega_1, \dots, \omega_d)$. The starting point is

$$\widehat{F}_\omega(t) = \frac{t^d}{\prod_{j=1}^d \sin(\pi \omega_j t)}. \quad (2.1)$$

The value at zero is filled in by continuity:

$$\widehat{F}_\omega(0) = \frac{1}{\pi^d \omega_1 \cdots \omega_d}.$$

The function is even. Write its Taylor series as

$$\widehat{F}_\omega(t) = \sum_{n \geq 0} b_n t^n, \quad \widetilde{F}_\omega(z) = \sum_{n \geq 0} n! b_n z^{-n-1}. \quad (2.2)$$

Thus our formal Borel map sends z^{-n-1} to $t^n/n!$. All odd b_n vanish. The positive poles, counted with multiplicity, are

$$\mathcal{P}_\omega = \{n/\omega_j : j = 1, \dots, d, n = 1, 2, \dots\}. \quad (2.3)$$

Different occurrences at the same location represent a higher-order pole, not separate simple residues. The numerator does not cancel any positive or negative pole.

For $0 < \theta < \pi/2$, define

$$F_\pm(z) = \int_0^{e^{\pm i\theta} \infty} e^{-zt} \widehat{F}_\omega(t) dt. \quad (2.4)$$

Initially the corresponding sufficient condition is $\operatorname{Re}(ze^{\pm i\theta}) > 0$. Integrals with different admissible angles on the same side agree on their common domains by contour deformation. For any z in the right half-plane, a sufficiently small positive θ satisfies both inequalities. These choices patch to two holomorphic functions on $\operatorname{Re} z > 0$. Throughout the article,

$$J_\omega(z) = F_-(z) - F_+(z), \quad S_\omega(z) = \frac{J_\omega(z)}{2\pi i}. \quad (2.5)$$

This sign convention matters in every residue formula below.

2.2 Uniform Gevrey estimates

Proposition 2.1 (Exact Gevrey order and off-ray realization). *The series $\widetilde{F}_\omega$ has Gevrey order one and no smaller nonnegative Gevrey order. Its Borel transform is meromorphic with a locally finite pole set. Fix*

$$0 < m_0 \leq \omega_j \leq M_0 < \infty \quad (j = 1, \dots, d). \quad (2.6)$$

There are constants $C, A > 0$, uniform on this box, such that $|n!b_n| \leq CA^n n!$. For fixed $\theta \in (0, \pi/2)$ there are uniform $C_\theta, A_\theta > 0$ for which, for $x > 0$ and $N \geq 0$,

$$\left| F_\pm(x) - \sum_{n=0}^{N-1} n!b_n x^{-n-1} \right| \leq C_\theta A_\theta^N N! x^{-N-1}. \quad (2.7)$$

Proof. Choose $0 < r < 1/M_0$. On $|t| \leq r$ the origin is removable and no other zero of a denominator occurs. Compactness of the period box gives a uniform bound there. Cauchy's inequality yields $|b_n| \leq Cr^{-n}$. For each fixed ω , the nearest poles are at $\pm 1/\max_j \omega_j$ and they are genuine. The Borel Taylor radius is therefore finite and equal to $1/\max_j \omega_j$. If $|n!b_n| \leq CB^n(n!)^s$ held with $s < 1$, then $|b_n|^{1/n} \leq CB(n!)^{(s-1)/n}$ up to an inessential factor

tending to one. The right side tends to zero, making the Borel transform entire. This is a contradiction.

On the ray $t = ue^{i\theta}$ one has

$$|\sin(\pi\omega_j t)|^2 = \sin^2(\pi\omega_j u \cos \theta) + \sinh^2(\pi\omega_j u \sin \theta).$$

Consequently $\hat{F}$ is uniformly bounded along that ray: it is regular near zero and decreases exponentially, up to the factor u^d , at infinity. The same holds below the axis. Let $r_0 = r/2$. Taylor's estimate on $u \leq r_0$ and the coefficient bound on $u \geq r_0$ imply a global ray estimate

$$\left| \hat{F}(ue^{\pm i\theta}) - \sum_{n < N} b_n(ue^{\pm i\theta})^n \right| \leq CA^N u^N.$$

For the second range, bound each polynomial term by $Cu^N r_0^{-N} (r_0/r)^n$ and sum the resulting geometric series; bound the function itself by its ray supremum. This proves the displayed estimate with constants independent of N . Integrating it against $e^{-xu \cos \theta}$ gives $CA^N N!/(x \cos \theta)^{N+1}$, which has the form (2.7). Integration of each Taylor monomial along the ray gives $n!x^{-n-1}$. The same decay estimates justify the contour patching in (2.4). $\square$

Remark 2.2 (What Gevrey regularity does not say). The coefficient bound concerns the Borel germ near the origin. It does not bound the residues at arbitrarily remote, nearly colliding singularities. No inference from (2.7) to absolute convergence of (1.1) will be made. The counterexample in Section 4 shows why such an inference would be false.

3 A separation-independent resonance-block theorem

3.1 The partition and explicit constants

Continue to assume the compact box (2.6). Set

$$\delta = \frac{1}{4dM_0}, \quad \rho = \frac{\delta}{4}. \quad (3.1)$$

List the positive pole occurrences in increasing order, keeping repeated nodes. Join two consecutive occurrences precisely when their gap is less than δ . The connected components are called *resonance blocks*. This is a fixed deterministic partition once the periods and the box are specified; it is not necessary to optimize the value of δ . For a block C , denote its ordered nodes by $a_1, \dots, a_m$, its size by $m = m_C$, and its minimum by a_C .

We use the deliberately conservative constants

$$\begin{aligned} B_0 &= \max \left\{ 1, \frac{2}{\pi m_0}, \frac{1}{m_0 \delta} \right\}, & L_0 &= 1 + \frac{1}{4M_0}, \\ K &= (L_0 B_0)^d, & D_0 &= d(\lceil M_0 \rceil + 2), \\ P_d(v) &= \sum_{m=1}^d \sum_{j=0}^{m-1} \frac{v^j}{j! \rho^{m-1-j}}. \end{aligned} \quad (3.2)$$

Sharper constants are possible. Their advantage here is that they require no number-theoretic information about the periods.

Lemma 3.1 (Finite block geometry). *Every block contains at most d occurrences and at most one occurrence from each lattice. Its diameter is less than $(d-1)\delta$. Distinct blocks are separated by at least δ . The number of block minima in any interval of length one is at most D_0 .*

Proof. Suppose $d + 1$ consecutive occurrences were connected. Their span would be less than $d\delta = 1/(4M_0)$. Two of the occurrences would come from the same one of the d lattices. Distinct nodes in that lattice are separated by at least $1/M_0$, a contradiction. The same argument excludes two occurrences from one lattice in any block. The diameter and separation claims follow from the construction. A lattice of spacing $1/\omega_j$ has at most $\lceil M_0 \rceil + 2$ nodes in any unit interval. Every block minimum is a node of one of the lattices, so summing this bound proves the final claim. $\square$

3.2 Removing the nearby poles before estimating

For each block define

$$H_C(t) = \widehat{F}_\omega(t) \prod_{\nu=1}^{m_C} (t - a_\nu). \quad (3.3)$$

All factors associated with repeated nodes are retained in the product. The point of (3.3) is that H_C , rather than any individual residue, has a uniform local bound.

Lemma 3.2 (Uniform regular-factor bounds). *The function H_C is holomorphic in a neighborhood of the closed tube*

$$\{t \in \mathbb{C} : \text{dist}(t, [a_1, a_m]) \leq 2\rho\}.$$

Throughout this tube,

$$|H_C(t)| \leq K(1 + a_C)^d. \quad (3.4)$$

For real $u \in [a_1, a_m]$ and every integer $k \geq 0$,

$$|H_C^{(k)}(u)| \leq k!K(1 + a_C)^d \rho^{-k}. \quad (3.5)$$

Proof. By Lemma 3.1, each lattice either contributes one canceled zero to the block or contributes none. For a contributing lattice, call its node a . On the stated tube,

$$|t - a| \leq (d - 1)\delta + 2\rho = (d - \tfrac{1}{2})\delta < \frac{1}{4M_0}.$$

After removing the harmless sign $(-1)^n$, $\sin(\pi\omega_j t) = \sin(\pi\omega_j(t - a))$. If $|w| \leq \pi/4$, the power series gives

$$\left| \frac{\sin w}{w} - 1 \right| \leq \frac{\sinh |w|}{|w|} - 1 < \frac{1}{4}.$$

It follows, with a slightly relaxed constant, that

$$\left| \frac{t - a}{\sin(\pi\omega_j t)} \right| \leq \frac{2}{\pi m_0}.$$

The canceled quotient is holomorphic, including at $t = a$.

For a noncontributing lattice, no node lies in the block interval, and all its positive nodes outside that interval are at least δ away. Zero and negative nodes are farther away still, because the first positive node of any lattice is at least $1/M_0$. Thus, throughout the tube,

$$\text{dist}(\text{Re } t, \omega_j^{-1}\mathbb{Z}) \geq \delta - 2\rho = \delta/2.$$

For real v , $|\sin(\pi v)| \geq 2 \text{dist}(v, \mathbb{Z})$. Together with $|\sin(u + iv)| \geq |\sin u|$, this yields $|\sin(\pi\omega_j t)| \geq m_0\delta$. Finally $|t| \leq a_C + 1/(4M_0) \leq L_0(1 + a_C)$. Multiplying the bounds for the d denominator factors and the numerator proves (3.4). Cauchy's estimate on a circle of radius ρ centered at u proves (3.5). $\square$

3.3 The cancellation identity

For a function f holomorphic near the nodes, its confluent divided difference is characterized by

$$[a_1, \dots, a_m]f = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(t)}{\prod_{\nu=1}^m (t - a_{\nu})} dt, \quad (3.6)$$

where Γ surrounds the nodes and no singularity of f . At distinct nodes this is

$$[a_1, \dots, a_m]f = \sum_{\nu=1}^m \frac{f(a_{\nu})}{\prod_{\mu \neq \nu} (a_{\nu} - a_{\mu})}. \quad (3.7)$$

At repeated nodes it is defined by continuous confluence; for m equal nodes it is $f^{(m-1)}(a)/(m-1)!$.

For real nodes we need the Hermite–Genocchi formula

$$[a_1, \dots, a_m]f = \int_{\Delta_{m-1}} f^{(m-1)}\left(\sum_{\nu=1}^m \lambda_{\nu} a_{\nu}\right) d\lambda. \quad (3.8)$$

Here $\lambda_{\nu} \geq 0$, $\sum \lambda_{\nu} = 1$, and the coordinate measure on the simplex has volume $1/(m-1)!$; it is not the induced Euclidean surface measure. For completeness, the formula for $m = 2$ is the fundamental theorem of calculus after a linear change of variable. For general m , integrate first along a simplex coordinate joining the first and last nodes. The fundamental theorem gives the difference of the two $(m-2)$ -simplex integrals divided by $a_m - a_1$. Induction gives the usual divided-difference recursion. Distinct nodes suffice for this argument; continuity of the compact simplex integral gives repeated nodes. Formula (3.6) satisfies the same recursion, by subtracting the two rational denominators, and the same initial case. This proves the identities, including the confluent case. See also [4].

Definition 3.3 (Block contribution). The contribution of a block is the finite sum over its distinct locations

$$S_C(z) = \sum_{a \in C \text{ distinct}} \text{Res}_{t=a} (e^{-zt} \hat{F}_{\omega}(t)). \quad (3.9)$$

Equivalently, without any separation denominator,

$$S_C(z) = [a_1, \dots, a_m](e^{-zt} H_C(t)). \quad (3.10)$$

Equation (3.10) follows immediately by substituting (3.3) into (3.6). It is an exact identity, not an asymptotic cancellation.

3.4 Normal convergence, Stokes equality, and the tail

Theorem 3.4 (Uniform resonance-block summation). *Under (2.6), for z with $x = \text{Re } z > 0$ and every block C ,*

$$|S_C(z)| \leq K(1 + a_C)^d e^{-a_C x} \sum_{j=0}^{m_C-1} \frac{|z|^j}{j! \rho^{m_C-1-j}}. \quad (3.11)$$

The series $\sum_C S_C(z)$ converges absolutely and locally uniformly on $x > 0$, and

$$F_-(z) - F_+(z) = 2\pi i \sum_C S_C(z). \quad (3.12)$$

For $T \geq 0$, retaining every block with $a_C < T$ gives the explicit bound

$$\left| S_{\omega}(z) - \sum_{a_C < T} S_C(z) \right| \leq \frac{K D_0 d! (T+2)^d P_d(|z|)}{(1 - e^{-x})^{d+1}} e^{-Tx}. \quad (3.13)$$

Every assertion and constant is uniform over the period box. The cutoff is block-complete: a retained block may contain a node slightly above T .

Proof. Put $m = m_C$ and $f(t) = e^{-zt}H_C(t)$. On the real block interval, Leibniz's rule and (3.5) give

$$\begin{aligned} |f^{(m-1)}(u)| &\leq K(1 + a_C)^d e^{-a_C x} \sum_{j=0}^{m-1} \binom{m-1}{j} |z|^j (m-1-j)! \rho^{-(m-1-j)} \\ &= (m-1)! K(1 + a_C)^d e^{-a_C x} \sum_{j=0}^{m-1} \frac{|z|^j}{j! \rho^{m-1-j}}. \end{aligned}$$

The simplex volume in (3.8) proves (3.11). In particular the bound depends on the size of a block, not the distance between any two of its nodes.

There are at most D_0 block minima in each interval $[T + k, T + k + 1)$. Hence

$$\begin{aligned} \sum_{a_C \geq T} |S_C(z)| &\leq K D_0 P_d(|z|) e^{-Tx} \sum_{k \geq 0} (T + k + 2)^d e^{-kx} \\ &\leq K D_0 P_d(|z|) (T + 2)^d e^{-Tx} \sum_{k \geq 0} (k + 1)^d e^{-kx}. \end{aligned}$$

The elementary generating function for $(k + 1)^d$ is an Eulerian polynomial divided by $(1 - q)^{d+1}$; the polynomial has nonnegative coefficients summing to $d!$. Thus $\sum_{k \geq 0} (k + 1)^d q^k \leq d! / (1 - q)^{d+1}$ for $0 < q < 1$. Alternatively this inequality follows by expanding the powers in rising factorials. This proves (3.13) and local uniform convergence.

It remains to identify the sum with the actual jump. Choose radii $R_\ell \rightarrow \infty$ at midpoints of the gaps between successive blocks. Each such radius is at least $\delta/2$ from every positive pole. For fixed z in the right half-plane choose a small $\theta > 0$ with $\operatorname{Re}(ze^{i\varphi}) \geq c > 0$ on $|\varphi| \leq \theta$. On the outer arc $t = R_\ell e^{i\varphi}$, the denominator is uniformly bounded away from zero when $|\operatorname{Im} t|$ is small, and grows exponentially when $|\operatorname{Im} t|$ is not small. More explicitly, for a fixed small $h > 0$, if $|\operatorname{Im} t| \leq h$, then $|R_\ell - \operatorname{Re} t| = O(h^2/R_\ell)$; for large ℓ the real part stays at least $\delta/4$ from every pole. If $|\operatorname{Im} t| \geq h$, use $|\sin(\pi \omega_j t)| \geq \sinh(\pi m_0 h)$. Thus $|\widehat{F}(t)| \leq C R_\ell^d$ everywhere on these arcs. Their integrals are $O(R_\ell^{d+1} e^{-c R_\ell})$ and tend to zero.

Apply the residue theorem to the contour that travels out along the lower ray, traverses the outer arc counterclockwise, and returns along the upper ray. All enclosed poles form complete blocks. There is no pole at zero. The limiting identity is (3.12), with the sign prescribed in (2.5). The same argument works locally uniformly in z . $\square$

Corollary 3.5 (Regularity across rational resonances). *The total jump is real-analytic in the positive periods, including at coincidences of any multiplicity at most d . Its values are independent of which sufficiently small admissible block threshold is used.*

Proof. For periods in a sufficiently small complex neighborhood of any fixed positive vector, fixed upper and lower rays avoid every zero of the corresponding denominators. Their imaginary parts remain bounded away from zero relative to the ray parameter outside a neighborhood of the origin. The integrals (2.4) therefore depend holomorphically on the periods, by locally uniform exponential domination. Their difference has the same property. Theorem 3.4 identifies every admissible partition with this one difference. Individual blocks can split or join when the partition changes; the total does not change. $\square$

Remark 3.6 (Not an unconditional rearrangement theorem). When the raw residue series is divergent, (3.12) is not a rearrangement of an already convergent series. It is a grouping prescription justified by contour exhaustion and a new absolute block estimate. Deleting one member of each close pair can destroy the conclusion. The finite grouping is part of the analytic representation.

4 The exact arithmetic obstruction for two lattices

4.1 Individual residues and the approximation exponent

Specialize to $d = 2$ and $(\omega_1, \omega_2) = (1, \alpha)$, where $\alpha > 0$ is irrational. All positive poles are simple. At the integer poles and the second-lattice poles the residues of the Borel kernel are

$$r_n = \frac{(-1)^n n^2}{\pi \sin(\pi \alpha n)}, \quad r'_m = \frac{(-1)^m (m/\alpha)^2}{\pi \alpha \sin(\pi m/\alpha)}. \quad (4.1)$$

The ungrouped candidate for the normalized jump is

$$\sum_{n \geq 1} r_n e^{-nz} + \sum_{m \geq 1} r'_m e^{-(m/\alpha)z}. \quad (4.2)$$

One may either ask that both series converge or order all their terms by increasing action. The absolute convergence criterion is the same, and the failure of the term test established below applies to either ordinary interpretation.

Let $\|u\| = \text{dist}(u, \mathbb{Z})$, and define the exponential approximation exponent

$$\beta(\alpha) = \limsup_{n \rightarrow \infty} \frac{1}{n} \log \frac{1}{\|n\alpha\|} \in [0, \infty]. \quad (4.3)$$

This exponent measures exponential, not merely polynomial, approximation. In particular, calling a number Liouville is not enough to determine whether its β is zero, finite positive, or infinite.

Lemma 4.1 (Reciprocal covariance). *For every positive irrational α ,*

$$\alpha\beta(1/\alpha) = \beta(\alpha). \quad (4.4)$$

Proof. Suppose first that $0 < b < \beta(\alpha)$, so there are infinitely many n with $\|n\alpha\| < e^{-bn}$. Let m be the nearest integer to $n\alpha$. Then $m/n \rightarrow \alpha$, and for these sufficiently large n , n is the nearest integer to m/α . Hence

$$\|m/\alpha\| = \frac{\|n\alpha\|}{\alpha}, \quad \limsup \frac{1}{m} \log \frac{1}{\|m/\alpha\|} \geq \frac{b}{\alpha}.$$

Letting b tend to $\beta(\alpha)$ proves $\beta(1/\alpha) \geq \beta(\alpha)/\alpha$ whenever the latter is positive, including the case of infinity. Interchanging α and $1/\alpha$ gives the reverse inequality. If one exponent is zero and the other were positive, the same positive-exponent argument in the reverse direction would give a contradiction. This covers all cases. $\square$

Theorem 4.2 (Exact ungrouped abscissa). *For irrational $\alpha > 0$, the raw residue series (4.2) converges absolutely when $\text{Re } z > \beta(\alpha)$. When $\text{Re } z < \beta(\alpha)$, its individual terms do not tend to zero. Consequently its abscissa of absolute convergence is exactly $\beta(\alpha)$, possibly infinite. No assertion is made here about convergence on the boundary $\text{Re } z = \beta(\alpha)$. In every case, the block representation of the same Stokes jump converges normally on the whole half-plane $\text{Re } z > 0$.*

Proof. For real u ,

$$2\|u\| \leq |\sin(\pi u)| \leq \pi\|u\|.$$

The factors n^2 and constant denominators in (4.1) do not contribute to logarithmic growth after division by n . Therefore

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log |r_n| = \beta(\alpha).$$

The root test proves absolute convergence of the first series for $\operatorname{Re} z > \beta(\alpha)$. If $\operatorname{Re} z < \beta(\alpha)$, choose a number strictly between them. Infinitely many coefficients grow faster than its exponential, and the corresponding terms fail to tend to zero. For the second lattice, the same root test in the index m gives the threshold $\operatorname{Re} z > \alpha\beta(1/\alpha)$, which is identical by Lemma 4.1. The terms of the first lattice already prove failure below the common threshold, regardless of how the two lattices are interleaved in an ordinary series. The final statement follows from Theorem 3.4 after choosing any compact positive box containing 1 and α . $\square$

4.2 A benign irrational and an extreme irrational

For $\alpha = \sqrt{2}$, let m be nearest to $n\sqrt{2}$. Since $2n^2 - m^2$ is a nonzero integer,

$$\|n\sqrt{2}\| = |n\sqrt{2} - m| = \frac{|2n^2 - m^2|}{n\sqrt{2} + m} \geq \frac{c}{n}$$

for a fixed $c > 0$ and all $n \geq 1$. Hence $\beta(\sqrt{2}) = 0$. The raw residue sum converges absolutely for every $\operatorname{Re} z > 0$, although block summation still improves conditioning near close poles.

In the opposite direction, define integers and a real number by

$$N_1 = 2, \quad N_{k+1} = 10^{2N_k}, \quad \alpha_* = 1 + \sum_{k \geq 1} 10^{-N_k}. \quad (4.5)$$

This decimal has infinitely many nonzero digits and unbounded gaps between them, so it is not eventually periodic and is irrational. It lies strictly between one and two. Put $q_k = 10^{N_k}$. The first k terms of $q_k \alpha_*$ are integers, while the tail satisfies

$$0 < \|q_k \alpha_*\| \leq 2 \cdot 10^{N_k - N_{k+1}}.$$

For sufficiently large k the tail is less than $1/2$, justifying the nearest-integer notation. Since $N_{k+1} = q_k^2$,

$$\frac{1}{q_k} \log \frac{1}{\|q_k \alpha_*\|} \geq \frac{(q_k^2 - N_k) \log 10 - \log 2}{q_k} \rightarrow \infty. \quad (4.6)$$

Thus $\beta(\alpha_*) = \infty$.

Corollary 4.3 (A simultaneous counterexample). *There is a formal series with all of the following properties:*

- (i) exact Gevrey order one and upper and lower Borel–Laplace sums;
- (ii) a meromorphic Borel transform having only simple poles, with no finite accumulation of poles;
- (iii) positive Stokes actions contained in a finitely generated, locally finite additive monoid;
- (iv) a raw individual-residue Stokes series that diverges at every z with $\operatorname{Re} z > 0$;
- (v) an absolutely normally convergent finite-block Stokes representation on that same half-plane, with an explicit tail estimate.

Proof. Take (2.1) with $d = 2$ and periods $(1, \alpha_*)$. Proposition 2.1 gives (i) and (ii). The support lies in $\{n + m/\alpha_* : n, m \in \mathbb{Z}_{\geq 0}\}$; its positive generators ensure local finiteness. Equations (4.6) and Theorem 4.2 prove (iv), while Theorem 3.4 proves (v). $\square$

The corollary isolates a limitation of a possible general principle, not of resurgence itself. Meromorphic continuation and Laplace summation still work; the pointwise sum of the individually resolved Stokes sectors is the operation that fails.

5 Finite-cut transseries remain valid

For a distinct positive pole a , let m_a be its multiplicity and write its principal part as

$$\widehat{F}_\omega(t) = \sum_{\ell=1}^{m_a} \frac{c_{a,-\ell}}{(t-a)^\ell} + O(1). \quad (5.1)$$

Then its exact residue contribution is

$$\text{Res}_{t=a}(e^{-zt}\widehat{F}_\omega(t)) = e^{-az}p_a(z), \quad p_a(z) = \sum_{\ell=1}^{m_a} c_{a,-\ell} \frac{(-z)^{\ell-1}}{(\ell-1)!}. \quad (5.2)$$

The polynomial has degree exactly $m_a - 1$, because the pole has that multiplicity and its highest principal coefficient is nonzero.

Theorem 5.1 (Finite-cut asymptotics and uniqueness). *Fix the positive periods. List their distinct positive poles increasingly as $a^{(1)} < a^{(2)} < \dots$. For each fixed $L \geq 0$ and real $x \rightarrow +\infty$,*

$$S_\omega(x) - \sum_{\ell=1}^L e^{-a^{(\ell)}x} p_{a^{(\ell)}}(x) = e^{-a^{(L+1)}x} p_{a^{(L+1)}}(x) + O\left(x^{d-1}e^{-b_Lx}\right) \quad (5.3)$$

for some $b_L > a^{(L+1)}$. The coefficients in this polynomial-exponential asymptotic expansion are unique. These statements hold even when the infinite individual-residue series diverges at every fixed positive x .

Proof. Only finitely many blocks meet the finite set of nodes through $a^{(L+1)}$. Include those entire blocks and, if necessary, the block containing the next node as well. A finite collection of individual residues accounts for all splitting of blocks caused by the desired cutoff. Within that finite collection, every node beyond $a^{(L+1)}$ has a strictly larger action. After it, the remaining complete blocks have a minimum action bounded strictly above $a^{(L+1)}$. Apply (3.13) at that latter fixed cutoff. Since $P_d(x) = O(x^{d-1})$ and $(1 - e^{-x})^{-d-1}$ is bounded for $x \geq 1$, this proves (5.3) for a suitable b_L .

For uniqueness, suppose two such expansions on locally finite positive action supports disagree. Their union is still locally finite and has a smallest action at which the polynomial coefficients differ. Subtract the expansions and multiply by the exponential of that action times x . The nonzero difference polynomial would be exponentially small times a fixed power of x , which is impossible as $x \rightarrow +\infty$. If the two supports include an action zero, the same argument starts there. Repeating at each finite cutoff proves uniqueness. $\square$

The constants in (5.3) may be enormous for a poorly approximable cutoff choice, because a fixed near-collision may be split. They are finite, which is all an asymptotic expansion at a fixed cutoff requires. In contrast, (3.13) supplies useful uniform constants when whole blocks are retained. This is the distinction between a formal or Poincaré-style expansion and a numerical evaluation scheme.

One should also distinguish finite-generation of the action monoid from finite-generation of a *function field*. Formal products have locally finite action support because sums of finitely many positive generators have only finitely many exponent tuples below any bound. This support fact alone does not give any analytic norm estimate for a product or an inverse. The block norms in Section 8 supply that missing information.

6 Coalescence and its universal scaled profile

6.1 An entire divided-difference profile

For $m \geq 1$, define

$$\Phi_m(\lambda_1, \dots, \lambda_m) = [\lambda_1, \dots, \lambda_m]e^{-u}. \quad (6.1)$$

At distinct nodes,

$$\Phi_m(\lambda_1, \dots, \lambda_m) = \sum_{j=1}^m \frac{e^{-\lambda_j}}{\prod_{k \neq j} (\lambda_j - \lambda_k)}. \quad (6.2)$$

The apparent singularities in (6.2) are removable. The representation

$$\Phi_m(\lambda) = (-1)^{m-1} \int_{\Delta_{m-1}} e^{-\sum \lambda_j v_j} dv \quad (6.3)$$

makes it entire and symmetric in its m complex arguments. In particular,

$$\Phi_m(\lambda, \dots, \lambda) = \frac{(-1)^{m-1}}{(m-1)!} e^{-\lambda}. \quad (6.4)$$

For real arguments its sign is $(-1)^{m-1}$. These properties are useful both for asymptotic analysis and for stable numerical evaluation.

Theorem 6.1 (Uniform x^{-1} action-coalescence law). *Let $x \rightarrow +\infty$, and suppose an m -node cluster has nodes*

$$a_j(x) = a + \frac{\lambda_j}{x} + O(x^{-2}), \quad j = 1, \dots, m, \quad (6.5)$$

uniformly for λ in a fixed compact subset of $\mathbb{R}^m$, with repeated nodes allowed. Suppose its regular factor H_x is holomorphic on a fixed neighborhood of a , its derivatives through order $m-1$ are uniformly bounded there, and $H_x(a) = H_0 + O(x^{-1})$, uniformly in those parameters. Its residue block then satisfies

$$S_C(x) = e^{-ax} x^{m-1} \left(H_0 \Phi_m(\lambda_1, \dots, \lambda_m) + O(x^{-1}) \right), \quad (6.6)$$

uniformly on the compact parameter set. The conclusion does not require a positive lower bound on $|\lambda_j - \lambda_k|$.

Proof. Let $\mu_j = x(a_j(x) - a) = \lambda_j + O(x^{-1})$. The affine change of variable $u = x(t - a)$ in a divided difference gives exactly

$$S_C(x) = e^{-ax} x^{m-1} [\mu_1, \dots, \mu_m] (e^{-u} H_x(a + u/x)).$$

On any fixed compact set containing the convex hulls of the μ_j , the function $H_x(a + u/x) - H_0$ is $O(x^{-1})$, as are its derivatives in u of orders up to $m-1$. This follows from the assumed derivative bounds and the chain rule; for order zero also use $H_x(a) - H_0 = O(x^{-1})$. The integral formula (3.8) therefore changes the last divided difference by $O(x^{-1})$ when this function is replaced by H_0 . Formula (6.3) is uniformly Lipschitz on compact sets, so replacing μ_j by λ_j creates another $O(x^{-1})$ error. No estimate has divided by a node separation. This proves the result. $\square$

Theorem 6.1 explains the polynomial factor generated by a multiple Borel pole. When m actions approach one another on the x^{-1} scale, the combined contribution is of order $x^{m-1} e^{-ax}$, with a finite profile interpolating all collision patterns. Exact coincidence is not a singular limit of the block.

For the sine-product model, the regular-factor hypotheses follow locally by canceling the selected lattice zeros before varying the periods. One must specify the collision under study and keep any other zeros out of the chosen neighborhood; arbitrary additional collisions would change m . Lemma 3.2 supplies uniform bounds for full blocks.

6.2 Two coincident lattices

For $d = 2$, suppose $a = n = m/\alpha$ is a common positive node. Expanding around $t = a + u$ gives

$$\widehat{F}_{(1,\alpha)}(a+u) = \frac{(-1)^{n+m}(a+u)^2}{\sin(\pi u)\sin(\pi\alpha u)}.$$

The coefficient of u^{-1} after multiplication by $e^{-x(a+u)}$ is

$$S_{\{a,a\}}(x) = \frac{(-1)^{n+m}}{\pi^2\alpha} e^{-ax}(2a - a^2x). \quad (6.7)$$

At $\alpha = 1$ every positive node is double. Summing the elementary geometric derivatives yields the exact expression

$$S_{(1,1)}(x) = \frac{1}{\pi^2} \left\{ \frac{2q}{(1-q)^2} - \frac{xq(1+q)}{(1-q)^3} \right\}, \quad q = e^{-x}. \quad (6.8)$$

It is valid for every $\operatorname{Re} x > 0$.

As a moving example take $\alpha = 1 + \lambda/x$ and the first two poles 1 and $1/\alpha$. Their scaled offsets are 0 and $-\lambda$, and $H_0 = 1/\pi^2$. Therefore

$$\frac{\pi^2 e^x}{x} S_{\{1,1/\alpha\}}(x) \longrightarrow -\frac{e^\lambda - 1}{\lambda}, \quad (6.9)$$

with value -1 at $\lambda = 0$. This is a uniform compact- λ limit, not just a statement for distinct limiting actions. At $\lambda = 0$, the finite- x expression on the left is exactly $-1 + 2/x$.

6.3 A triple-pole check

For $(\omega_1, \omega_2, \omega_3) = (1, 1, 1)$,

$$\frac{1}{\sin^3(\pi u)} = \frac{1}{\pi^3 u^3} \left(1 + \frac{\pi^2 u^2}{2} + O(u^4) \right).$$

Extracting the residue at the integer n gives

$$S_{\{n,n,n\}}(x) = \frac{(-1)^n e^{-nx}}{\pi^3} \left(3n - 3n^2 x + \frac{n^3}{2} (x^2 + \pi^2) \right). \quad (6.10)$$

The leading term has the $x^2 e^{-nx}$ behavior predicted by (6.6) and $\Phi_3(0, 0, 0) = 1/2$. This example also shows why limiting residues must be taken as higher-order residues rather than by substituting a resonant parameter into either simple-pole formula.

7 Rational-exponential compression and its converse

Definition 7.1. A function on a positive half-line has a *finite single-base rational-exponential representation* if for some $h > 0$, some integer $N \geq 0$, and rational functions R_k holomorphic at the origin, it can be written

$$S(x) = \sum_{k=0}^N x^k R_k(e^{-hx}). \quad (7.1)$$

This is a specific compression class. It does not include arbitrary rational functions of several independent exponential bases, nor every special-function representation.

Theorem 7.2 (Exact two-lattice compression criterion). *For the two-lattice kernel $\hat{F}_{(1,\alpha)}$, the normalized jump $S_{(1,\alpha)}$ has a representation (7.1) if and only if α is rational. If $\alpha = p/q$ in lowest terms with $p, q > 0$, one may take $h = 1/p$ and $N \leq 1$. More generally, for d commensurate positive periods one may take $N \leq d - 1$ with a suitable h .*

Proof. Suppose first that the lattice spacings are commensurate. There are $h > 0$ and positive integers r_j with $1/\omega_j = hr_j$. All nodes have the form $h\ell$, and the multiplicity at that node is the number of indices for which r_j divides ℓ . This pattern is periodic in ℓ . Expand each denominator around $t = h\ell + u$. Whether its sine vanishes, and every coefficient needed from its Laurent or Taylor expansion, depend periodically on ℓ , with a common period dividing a multiple of $2\text{lcm}(r_1, \dots, r_d)$. The numerator $(h\ell + u)^d$ is polynomial in ℓ . The residue polynomial (5.2) thus has the form

$$p_{h\ell}(x) = \sum_{k=0}^{d-1} x^k A_k(\ell),$$

where each A_k is polynomial in ℓ separately on each of finitely many residue classes. Terms at locations that are not poles are set to zero. For $Q = e^{-hx}$, every series $\sum_{\ell \geq 1} A_k(\ell) Q^\ell$ is rational in Q : split into residue classes and apply powers of $Q\partial_Q$ to geometric series. The coefficients have only polynomial growth, so these sums converge absolutely when $|Q| < 1$. This proves (7.1) with $N \leq d - 1$ and rational functions holomorphic at zero. For $(1, p/q)$ the spacings are 1 and q/p , so $h = 1/p$ works and $d - 1 = 1$.

Conversely suppose α is irrational and (7.1) holds. Expanding the rational functions at zero gives a convergent polynomial-exponential expansion supported on $\{h\ell : \ell \geq 0\}$. The zero-action polynomial vanishes because $S_{(1,\alpha)}(x)$ is exponentially small. Every pole in the actual kernel is simple and has a nonzero residue. By Theorem 5.1, its asymptotic action expansion is unique. In particular the actions 1 and $1/\alpha$ must both be positive integer multiples of h . Their ratio would then be rational, contradicting the irrationality of α . This argument uses finite-cut asymptotics, not convergence of the raw infinite series, so it also applies when $\beta(\alpha) = \infty$. $\square$

The theorem gives a precise contrast. The total jump varies analytically through rational periods by Corollary 3.5, yet admitting a finite single-base representation is an arithmetic property attained exactly on rational two-lattice ratios. Smooth dependence of a function on a parameter does not imply uniform finite symbolic compressibility.

8 Nonlinear inversion in the block norm

8.1 The analytic space that survives the small divisors

The appropriate norm is taken after finite cancellation. Let D be a closed disk in the right half-plane. Theorem 3.4 gives

$$\sum_C \sup_{z \in D} |S_C(z)| < \infty. \quad (8.1)$$

The corresponding sum over individual residues need not be finite. For the example α_* , it is infinite on every such disk with nonempty interior, because even evaluation at any one point produces terms that do not tend to zero.

We shall call (8.1) the *block norm*. This terminology refers to a chosen representation in a Banach space of holomorphic functions on a disk. It does not assert the construction of a new full field of formal transseries. In particular, different decompositions can have different norms even when their sums are the same function.

8.2 A branch certificate and all-order inverse transport

Theorem 8.1 (Blockwise inverse transport and two truncation errors). *Let ϕ be univalent and holomorphic on a neighborhood of $\overline{D}(y_0, R)$, let $z_0 = \phi(y_0)$, and let $G = \phi^{-1}$ be the corresponding local core. Suppose*

$$j(y) = J(\phi(y)) = \sum_C j_C(y), \quad B = \sum_C \sup_{|y-y_0| \leq R} |j_C(y)| < \infty, \quad (8.2)$$

where the j_C are holomorphic on a common neighborhood of the closed disk, and the sum is holomorphic there or on an open neighborhood of the disk after an arbitrarily small reduction of R . Put

$$M = \sup_{|y-y_0| \leq R} |\phi'(y)|.$$

If $r = |\lambda|B/R < 1$, the equation

$$G(z_\lambda) + \lambda J(z_\lambda) = y_0 \quad (8.3)$$

has a unique solution in $\phi(D(y_0, R))$. It depends holomorphically on λ and satisfies

$$z_\lambda - z_0 = \sum_{n \geq 1} \frac{(-\lambda)^n}{n!} \partial_y^{n-1} (\phi'(y) j(y)^n) \Big|_{y=y_0}. \quad (8.4)$$

Expanding each j^n over ordered n -tuples of blocks is absolutely convergent. The sum of absolute values of all degree- n contributions is at most MRr^n/n , and the tail beyond degree $N \geq 0$ is bounded by

$$\frac{MR}{N+1} \frac{r^{N+1}}{1-r}. \quad (8.5)$$

For a subfamily of retained blocks let j_T be its sum, and assume

$$\Delta_T = \sum_{C \text{ omitted}} \sup_{|y-y_0| \leq R} |j_C(y)|. \quad (8.6)$$

The full inverse and the inverse formed with j_T differ by at most

$$\frac{M|\lambda|\Delta_T}{1-r}. \quad (8.7)$$

Thus truncating both the blocks and the amplitude degree gives a total error bounded by the sum of (8.7) and (8.5). If $B = 0$, all statements have the trivial interpretation $z_\lambda = z_0$.

Proof. Use the core coordinate $y = G(z)$. The equation becomes $y + \lambda j(y) = y_0$. On $|y - y_0| = R$, its perturbation has modulus at most $|\lambda|B < R$. Rouché's theorem gives exactly one zero in the disk, counted with multiplicity. It is therefore simple. The implicit function theorem and uniqueness give a single holomorphic branch in $|\lambda| < R/B$.

The argument principle, applied with the holomorphic weight $\phi(y)$, expresses the difference of the roots after applying ϕ as

$$\frac{1}{2\pi i} \int_{|y-y_0|=R} \phi(y) \left(\frac{1 + \lambda j'(y)}{y - y_0 + \lambda j(y)} - \frac{1}{y - y_0} \right) dy.$$

The logarithm $\log(1 + \lambda j(y)/(y - y_0))$ is single-valued on a thin annulus containing the contour, by its uniformly convergent power series. Integration by parts changes the preceding expression to

$$-\frac{1}{2\pi i} \int_{|y-y_0|=R} \phi'(y) \log \left(1 + \frac{\lambda j(y)}{y - y_0} \right) dy. \quad (8.8)$$

Expanding the logarithm and applying Cauchy's formula proves (8.4).

For the absolute estimate, retain the contour form of each homogeneous term. The sum of the moduli of all ordered products of block factors is bounded on the contour by $(\sum_C \|j_C\|)^n = B^n$. The contour length is $2\pi R$, the core denominator has modulus R^n , and the logarithmic coefficient is $1/n$. Consequently the total degree- n bound is MR^n/n . It justifies every interchange of sums, derivatives, and contour integrals. Summing for $n \geq N+1$ and using $1/n \leq 1/(N+1)$ proves (8.5).

For the second statement use (8.8) for j and j_T . Along the segment between their two contour values the arguments of the logarithm remain in $|w-1| \leq r$. Hence the difference of the two logarithms has modulus at most $|\lambda|\Delta_T/(R(1-r))$. Multiplying by the contour length and $M/(2\pi)$ gives (8.7). Both equations have a unique root by the same strict boundary bound. This finishes the proof. $\square$

The contour proof is a form of Lagrange inversion, and the corresponding weighted-family result already appears in the repository [1]. We included the proof to make the logical dependence and constants explicit. The new input is the finite-block estimate that verifies the norm hypothesis in examples where the individual-sector version fails.

8.3 Explicit certificates for the sine-product family

Take $J = J_\omega$ from (2.5) and $j_C(y) = 2\pi i S_C(\phi(y))$. Suppose the core image satisfies

$$\operatorname{Re} \phi(y) \geq x_* > 0, \quad |\phi(y)| \leq V \quad (|y - y_0| \leq R). \quad (8.9)$$

Since every block minimum is at least $1/M_0$, the proof of Theorem 3.4 gives the directly evaluable certificate

$$B \leq \frac{2\pi K D_0 d! (2 + M_0^{-1})^d P_d(V)}{(1 - e^{-x_*})^{d+1}} e^{-x_*/M_0}. \quad (8.10)$$

For the block-complete cutoff $a_C < T$,

$$\Delta_T \leq \frac{2\pi K D_0 d! (T + 2)^d P_d(V)}{(1 - e^{-x_*})^{d+1}} e^{-Tx_*}. \quad (8.11)$$

These are bounds on sums of suprema, not merely bounds on the absolute value of the total perturbation. This distinction is essential to absolute mixed-block convergence.

Equations (8.10), (8.5), and (8.11) provide two independent accuracy controls. Increasing T reduces the unresolved action tail; increasing N reduces the nonlinear amplitude tail. Neither control requires a lower bound on an internal action separation or an estimate of $\beta(\alpha)$.

Corollary 8.2 (A genuine extension beyond individual-sector inversion). *For the irrational α_* in (4.5), the block-indexed inverse expansion of Theorem 8.1 is applicable whenever its branch certificate holds. In contrast, an inverse expansion resolved into the raw individual residues fails the term test already in its first homogeneous degree at every base point z_0 with $\operatorname{Re} z_0 > 0$.*

Proof. The block norm is finite by (8.10). The degree-one coefficient is

$$-\lambda \phi'(y_0) J(z_0). \quad (8.12)$$

Expanding $J(z_0)$ into individual residues yields terms equal to the raw terms in Theorem 4.2, multiplied by the nonzero constant $-2\pi i \lambda \phi'(y_0)$. For nonzero λ they fail to tend to zero. In block form the same degree-one coefficient is an absolutely convergent sum. Higher-order rearrangements cannot repair failure of absolute convergence at the first homogeneous degree. $\square$

8.4 Formal action preservation with analytic realization

Let $\mathcal{A}$ be the distinct positive pole set, let $a_* = \min \mathcal{A}$, and put $q_a(x) = 2\pi i p_a(x)$, with p_a from (5.2). The formally resolved jump is $\tilde{J}(x) = \sum_{a \in \mathcal{A}} q_a(x) e^{-ax}$.

Theorem 8.3 (An explicit finite formula at each inverse action). *For the identity core, the formal inverse of $x \mapsto x + \lambda \tilde{J}(x)$ is well-defined on the additive action monoid $\langle \mathcal{A} \rangle$. Its correction has coefficients*

$$\tilde{Z}_\lambda(x) - x = \sum_{A \in \langle \mathcal{A} \rangle \setminus \{0\}} Q_A(x; \lambda) e^{-Ax}, \quad (8.13)$$

$$Q_A(x; \lambda) = \sum_{1 \leq n \leq A/a_*} \frac{(-\lambda)^n}{n!} \sum_{\substack{a_1, \dots, a_n \in \mathcal{A} \\ a_1 + \dots + a_n = A}} (\partial_x - A)^{n-1} \prod_{j=1}^n q_{a_j}(x). \quad (8.14)$$

Every sum defining Q_A is finite. Thus Q_A is polynomial in x and λ , and no new action outside the additive input monoid is needed. For fixed $\lambda \in \mathbb{C}$, the actual inverse near large positive x of $z \mapsto z + \lambda J_\omega(z)$ has this expansion at every fixed finite action cutoff. Its block-indexed amplitude series is convergent for all sufficiently large x , even in the counterexample of Corollary 4.3.

Proof. The monoid generated by the d positive spacings is locally finite, and $\mathcal{A}$ is contained in it. A product of n positive-action terms has action at least na_* . Hence only $n \leq A/a_*$ and finitely many action tuples can contribute at action A . Differentiation does not change an action and satisfies

$$\partial_x^{n-1} (e^{-Ax} P(x)) = e^{-Ax} (\partial_x - A)^{n-1} P(x).$$

The identity-core case of the Lagrange formula therefore gives (8.14). It can also be verified in each finite action quotient: substitution by a positive-action displacement raises its filtration, so the usual recursive inverse is unique, and the finite Lagrange identities hold at each such quotient. This proves formal existence and uniqueness without an analytic convergence assumption on $\tilde{J}$.

For the analytic assertion take a fixed-radius disk about real x and apply Theorem 8.1 with $\phi(y) = y$. Its block norm is $O(x^{d-1} e^{-a_* x})$; a constant factor from the disk radius is harmless. It tends to zero, so every fixed λ eventually satisfies the branch certificate. Fix an action bound A_0 . Choose N with $(N+1)a_* > A_0$. The amplitude tail (8.5) is then bounded by a fixed power of x times $e^{-(N+1)a_* x}$ and is smaller than every retained term at action at most A_0 . For each of the finitely many degrees $n \leq N$, retain finitely many whole jump blocks through an action strictly above A_0 . The omitted jump tail is exponentially smaller, uniformly on a slightly larger fixed-radius disk, by (3.13). Cauchy's formula preserves this exponential margin after the finitely many derivatives. The retained finite products give exactly (8.14) through A_0 . This proves finite-cut realization. It does not assert convergence of the infinite raw action sum at fixed x . $\square$

8.5 Actual lateral maps

For a fixed scalar ε , consider

$$T_\pm(z) = z + \varepsilon F_\pm(z). \quad (8.15)$$

On a fixed-width neighborhood of a sufficiently large positive real point, the ray estimates give $F_\pm(z) = O(z^{-1})$ and $F'_\pm(z) = O(z^{-2})$. These follow either by differentiating the convergent ray integral or by its first Taylor-remainder estimate on a slightly larger disk. Thus T_+ is a small derivative perturbation of the identity and has a local inverse core ϕ on a disk of fixed positive radius, after moving the base point sufficiently far right.

Set $G = T_+$ and $\lambda = \varepsilon$ in Theorem 8.1. Since $T_- = T_+ + \varepsilon J$, its solution is $T_-^{-1}(y_0)$ on the certified branch. This realizes the inverse Stokes transport as an actual comparison of lateral

inverse maps, not merely as a formal coefficient calculation. The bound (8.10) is exponentially small as the base point moves right in a bounded imaginary strip, so the branch condition eventually holds uniformly on a compact period box. No global univalence or a complete description of the inverse's Borel singularities is being asserted.

9 A more general local criterion

The sine product is useful because it supplies all geometric and analytic constants explicitly. The cancellation step itself does not depend on trigonometric functions.

Proposition 9.1 (Bounded-cluster meromorphic kernels). *Let $\hat{H}$ be meromorphic near the positive axis and regular at zero. Suppose its positive pole occurrences admit consecutive blocks C satisfying the following conditions. Their multiplicities obey $m_C \leq d$, their real nodes lie in intervals I_C , and the number of block minima in every unit interval is at most D . For a fixed $\rho > 0$, the functions*

$$H_C(t) = \hat{H}(t) \prod_{a \in C} (t - a)$$

are holomorphic on the 2ρ tubes around I_C and satisfy $|H_C(t)| \leq K(1 + a_C)^p$, where p is a nonnegative integer. Then the associated residue blocks obey

$$|S_C(z)| \leq K(1 + a_C)^p e^{-a_C \operatorname{Re} z} \sum_{j=0}^{m_C-1} \frac{|z|^j}{j! \rho^{m_C-1-j}}. \quad (9.1)$$

They have a normally convergent sum on the right half-plane and the tail bound

$$\sum_{a_C \geq T} |S_C(z)| \leq \frac{K D p! (T + 2)^p P_d(|z|)}{(1 - e^{-\operatorname{Re} z})^{p+1}} e^{-T \operatorname{Re} z}. \quad (9.2)$$

If upper and lower Laplace integrals exist and a block-complete sequence of closing contours encloses exactly the specified positive poles, no other singularities, and has vanishing outer integrals, this sum is their Stokes difference divided by $2\pi i$.

Proof. Cauchy's derivative bound on each interval and the confluent Hermite–Genocchi formula prove (9.1) exactly as in Theorem 3.4. Binning minima into unit intervals and using $\sum_{k \geq 0} (k + 1)^p q^k \leq p! / (1 - q)^{p+1}$ proves (9.2), including $p = 0$. The stated contour hypothesis then gives the residue identity by the residue theorem. Each hypothesis is used explicitly; no conclusion about lateral Laplace sums follows from the local block bound alone without the contour conditions. $\square$

This criterion separates two tasks in applications. One must first find clusters of bounded multiplicity and bound their pole-canceled regular factors. One must then identify their normal sum with the intended analytic realization. In the sine-product model these tasks were proved in Lemmas 3.1–3.2 and Theorem 3.4, respectively. Holomorphic multipliers with suitable polynomial tube bounds can be accommodated by changing p and K . Arbitrary multipliers, arbitrary pole densities, or unbounded cluster cardinality require new estimates and are not covered for free.

10 Stable evaluation and reproducible checks

10.1 A cancellation-free two-pole formula

Suppose two distinct near-colliding nodes are $a = n$ and $b = m/\alpha$, and put

$$\epsilon = b - a, \quad \sigma = (-1)^{n+m}.$$

For this finite pair, elementary sine identities transform the residue sum into

$$S_{\{a,b\}}(x) = \frac{\sigma e^{-ax}}{\pi^2 \alpha} \left\{ \frac{2a + \epsilon - xb^2 \operatorname{exprel}(-x\epsilon)}{\operatorname{sinc}(\pi\epsilon)} + a^2 K_\alpha(\epsilon) \right\}, \quad (10.1)$$

where

$$\begin{aligned} \operatorname{sinc} u &= \frac{\sin u}{u}, & \operatorname{sinc} 0 &= 1, \\ \operatorname{exprel} v &= \frac{e^v - 1}{v}, & \operatorname{exprel} 0 &= 1, \\ K_\alpha(\epsilon) &= \frac{\operatorname{sinc}(\pi\epsilon)^{-1} - \operatorname{sinc}(\pi\alpha\epsilon)^{-1}}{\epsilon}, & K_\alpha(0) &= 0. \end{aligned} \quad (10.2)$$

In numerical code, `exprel` is evaluated through `expm1`, not by subtracting one from an exponential near one.

To verify (10.1), use $\alpha a = m - \alpha\epsilon$ and $b = n + \epsilon$ in (4.1). The two contributions become

$$\frac{\sigma}{\pi^2 \alpha \epsilon} \left(\frac{b^2 e^{-bx}}{\operatorname{sinc}(\pi\epsilon)} - \frac{a^2 e^{-ax}}{\operatorname{sinc}(\pi\alpha\epsilon)} \right).$$

Now split the difference into the change in $b^2 e^{-bx}$ and the change in reciprocal `sinc`. The first difference divided by ϵ is exactly $e^{-ax}(2a + \epsilon - xb^2 \operatorname{exprel}(-x\epsilon))$. This proves the formula. At $\epsilon = 0$ it becomes the exact higher-order residue (6.7).

The remaining difference in K_α should also not be evaluated by subtraction when ϵ is tiny. Its convergent series is

$$K_\alpha(\epsilon) = \sum_{k \geq 1} c_k \pi^{2k} (1 - \alpha^{2k}) \epsilon^{2k-1}, \quad c_k = \frac{(-1)^{k+1} (2^{2k} - 2) B_{2k}}{(2k)!}, \quad (10.3)$$

valid for $\max(1, \alpha)|\epsilon| < 1$. Here B_{2k} are Bernoulli numbers, and c_k is the coefficient of u^{2k} in $u/\sin u$. Near $\alpha = 1$, the factor $1 - \alpha^{2k}$ can be evaluated as $-\operatorname{expm1}(2k \log \alpha)$. This formula exposes the smooth limit before finite-precision arithmetic is applied.

For larger blocks one can use the simplex integral, a confluent divided-difference algorithm with Taylor data, or contour quadrature around the whole block. The stable two-pole formula is an explicit implementation for $d = 2$, not a claim that ordinary divided-difference recursion is uniformly stable at arbitrarily close nodes.

10.2 An explicit action budget

For a target normalized-jump error $\eta > 0$ at real $x > 0$, a sufficient cutoff is any T satisfying

$$Tx \geq \log \frac{K D_0 d! P_d(x)}{\eta (1 - e^{-x})^{d+1}} + d \log(T + 2). \quad (10.4)$$

The right side grows only logarithmically in T , so a finite value is always available. Its sufficiency is proved by (3.13); evaluating the displayed bound using ordinary floating-point arithmetic is not by itself an outward-rounded certificate. A rigorous numerical implementation would evaluate it and all retained blocks with interval arithmetic.

There are $O_{d,M_0}(T + 1)$ pole occurrences below T and at most d per block. Merging the d ordered lattice lists and evaluating each block gives an action-linear enumeration cost for fixed d , before accounting for arithmetic precision and special-function evaluation. A claim of uniform bit complexity would require a representation of the periods and certified comparisons of the gaps. Arbitrary exact real inputs do not provide such an algorithm automatically.

10.3 Executed verification

The accompanying program `verify.py` uses SymPy 1.14.0 and mpmath 1.3.0 under Python 3.13.5. The recorded run contains **36 exact symbolic assertions and 15 numerical assertions**, all passing. The symbolic checks cover the first Borel coefficients, vanishing odd coefficients, double and triple residues, nine cosecant–Bernoulli coefficients, geometric-series identities, twelve inverse coefficients for an exactly solvable nonlinear core, and divided differences at three sets of rational nodes. The numerical checks compare upper-ray Laplace integrals with block residue sums and test the stable pair formula near resonance.

For $d = 2$ on the period box $[1/2, 2]$, the chosen constants become

$$\delta = \frac{1}{16}, \quad \rho = \frac{1}{64}, \quad K = 1296, \quad D_0 = 8, \quad P_2(v) = 65 + v. \quad (10.5)$$

At $x = 3$ and $T = 25$, (3.13) evaluates to approximately $3.2093159381 \times 10^{-24}$. Direct comparisons at 90 decimal digits give the following discrepancies. The independently evaluated contour quantity is $-\operatorname{Im} F_+(x)/\pi$, because the two lateral integrals are complex conjugates for these real parameters.

Period ratio α	Absolute contour–block discrepancy
1	5.2353×10^{-31}
$\sqrt{2}$	5.1155×10^{-31}
$3/2$	3.9773×10^{-31}
$3/2 + 10^{-30}$	3.9773×10^{-31}

These discrepancies are consistent with the proved bound, which is intentionally conservative. For $\alpha = 1$ the contour was also compared with the exact rational expression (6.8).

The stable pair was tested at $\alpha = 1 + 10^{-k}$ for $k \in \{4, 10, 25, 40, 60\}$, with 200-digit reference calculations. A separate 50-digit demonstration at $\alpha = 1 + 10^{-40}$ gave relative error about 6.43×10^{28} for direct evaluation of the two raw sine residues, but about 2.88×10^{-51} for (10.1). The disastrous raw error reflects both evaluation near zeros of sine and subtraction of very large contributions. It is a demonstration of one unstable formula at a specified precision, not a universal lower bound on required precision.

Table 1 displays the scaled pair in (6.9). It is generated by the supplied code. At the smallest values of x , a selected pair need not yet belong to a single block for the fixed threshold (10.5); the two-pole formula itself is still exact. For sufficiently large x it is a single resonance block, as required by the limiting interpretation.

λ	$x = 20$	$x = 80$	$x = 320$	$x = 1280$	Limit
-2	-0.4400485	-0.43281731	-0.43236275	-0.43233426	-0.43233236
-0.5	-0.74088243	-0.77553763	-0.78409388	-0.7862278	-0.78693868
0	-0.9	-0.975	-0.99375	-0.9984375	-1.0
0.5	-1.1014891	-1.2465872	-1.2846042	-1.294225	-1.2974425
2	-2.0723416	-2.8518498	-3.1039264	-3.1715488	-3.194528

Table 1: Scaled first-pair contribution for $\alpha = 1 + \lambda/x$ and its uniform coalescence limit.

Recorded numerical values, package versions, and assertion counts are stored in `results/verification.json`. These computations do not use outward-rounded intervals, do not prove the infinite tail estimates, and do not verify all theorem hypotheses for arbitrary input periods. The mathematical proofs, not a finite number of experiments, establish the claims for arbitrary positive periods.

11 Further research questions and proposed projects

The following questions are proposed research tasks, not claims that specific long-standing conjectures have been settled here. Several may have answers in specialized small-divisor or special-function literature; the first step of each project should include a targeted priority search. The sharp boundaries of the proved results are indicated in the questions.

1. The boundary of the raw-series half-plane

Theorem 4.2 leaves $\operatorname{Re} z = \beta(\alpha)$ open. Determine necessary and sufficient criteria for absolute and conditional convergence there in terms of the continued-fraction denominators of α . The polynomial factor n^2 matters on the boundary and is deliberately invisible to the exponent β . A useful result would distinguish the two separately indexed series from the increasing-action ordering and would include explicit examples at every attainable convergence type.

2. Sharp absolute bounds for blocks

The constants in (3.2) sacrifice sharpness for uniformity. For $d = 2$, optimize the pair and singleton estimates and identify the best leading tail constant for a fixed compact period interval. The exact profile (6.3) should permit substantially smaller bounds than replacing every derivative by a common Cauchy majorant. A first concrete target is a tail estimate whose leading power of T is sharp both near and away from rational ratios.

3. All-order moving-resonance expansions

Theorem 6.1 proves the leading profile with a uniform $O(x^{-1})$ correction. Derive an all-order expansion when the periods have complete asymptotic expansions in $1/x$, expressing every term through derivatives of Φ_m and Taylor coefficients of the regular factor. The algebraic calculation is accessible, but a useful theorem must give a uniform remainder on compact sets crossing every partial collision stratum. Uniformity on growing parameter windows would then connect this expansion with well-separated residue asymptotics.

4. Unbounded multiplicity and countably many lattices

A finite d bounds the number of nodes in a block. For infinitely many lattices that argument fails: clusters can contain arbitrarily many nodes, while the factors x^{m-1} , derivative estimates, and simplex dimensions all change. Find a weighted condition balancing cluster cardinality, regular-factor norms, and action growth that yields a nontrivial countable-lattice analogue of (9.2). This is a genuine extension beyond the hypotheses of Proposition 9.1, not an immediate consequence of the finite-dimensional result.

5. Nonmeromorphic singularities in a resonance block

Replace simple and higher-order poles by algebraic or logarithmic Borel singularities. Seek a block analogue of divided differences in which Hankel-contour contributions remain uniformly controlled as their branch points collide. The correct collision profile may involve beta integrals rather than the simplex exponential in (6.3). One must also specify branch choices and allowed continuation paths, which are absent in the meromorphic model.

6. Quantitative resurgence of the nonlinear inverse

Theorem 8.3 provides formal action preservation and an analytic realization near the positive axis. It does not give an explicit continuation theorem for the Borel transform of the inverse on

all relevant sheets. Determine the singularity types, local coefficients, and continuation bounds created by nonlinear convolution when the primitive Borel poles have arbitrarily close neighbors. Established resurgent closure theorems provide a framework; the target is quantitative control uniform in the period parameters, rather than closure in name alone.

7. Multiple exponential bases and differential algebra

Theorem 7.2 classifies one-base rational-exponential representations. Determine what changes if a finite number of independent bases is allowed, or if one permits algebraic, differential-algebraic, or multiple-sine special functions. In particular, a useful obstruction should not merely restate the irrational ratio of two generators, since that obstruction disappears when two bases are admitted.

8. Certified and precision-adaptive computation

Implement interval-arithmetic versions of (10.1) and (3.13). A full certificate must cover input period uncertainty, pole ordering, grouping decisions, local residue evaluation, and the unresolved tail. If an uncertain gap straddles the grouping threshold, merge a larger certified cluster rather than guessing its order, and track the resulting multiplicity bound. Analyze bit complexity for algebraic periods and for computable periods supplied with certified approximants.

9. Other exact special-function realizations

The sine-product kernels are close in structure to kernels appearing in multiple-gamma, multiple-sine, and quantum-dilogarithm integral formulas. Identify exact parameter and differentiation correspondences rather than relying on formal resemblance. Then compare the resulting block sums with known product and modular representations, including the normalization polynomials and lateral sign conventions. This may reveal which parts of the present quantitative package are refinements of established special-function identities.

10. Negative and genuinely complex actions

Our right-half-plane norm exploits positive real actions. Study clusters on several complex rays and the geometry of the common Laplace domains. Products can create cancellation of complex actions, including action zero, so positivity-based finite-cut arguments may fail. A useful extension would give a cone condition that simultaneously controls cluster summation, nonlinear products, and inverse support.

11. Optimal groupings and representation invariants

Different admissible partitions produce the same jump but different block norms. Is there a canonical or near-optimal grouping for numerical work or for a specified inverse disk? Determine which information is invariant under finite refinement and coarsening, and whether a representation-independent analytic norm can retain the useful separation-free estimates. The goal is a useful analytic structure, not merely a new name for an existing contour integral.

12. A focused Lean formalization

Formalize the finite geometric and algebraic core first: the pigeonhole bound on block size, confluent residue identities for double and triple poles, and the finite action-coefficient formula (8.14). Next connect a finite divided-difference API to an analytic norm estimate. The full theorem additionally requires complex contour integration, a normally convergent infinite block

sum, and a branch-sensitive inverse construction. These layers should be reported separately; formalizing the finite algebra alone would not formalize the Stokes theorem.

12 Conclusion

The distinction established by the model can be summarized without a summability slogan. The Borel germ is uniformly Gevrey-one. Its positive singularities have a locally finite, finitely generated action support. Their individual residue amplitudes can nevertheless grow so quickly that no positive value of the large variable makes the raw Stokes sum converge. The failure is arithmetic and its two-lattice abscissa is exactly $\beta(\alpha)$.

Canceling nearby poles before estimating gives a different conclusion. Every resonance block has bounded cardinality, a controlled holomorphic regular factor, and an exact divided-difference representation. The resulting block sum converges normally, equals the lateral jump, and varies regularly across rational resonances. It admits both an explicit action-cutoff error and an analytic inverse expansion with a separate amplitude-degree error. At every fixed formal action cutoff the original polynomial-exponential transseries remains meaningful, even in examples where its raw infinite evaluation fails everywhere.

Thus the natural analytic datum for this class is not the support alone, nor the list of individual residues with an imposed absolute-summability condition. It is the support together with a finite cancellation structure and a norm on its blocks. This is the specific extension to the repository's transseries and inverse-transport program proved here.

A Proof dependencies and scope audit

Result	Main ingredients	Boundary of the claim
Gevrey realization, Proposition 2.1	Cauchy coefficients, nonreal-ray sine bounds, Laplace remainder	Does not control individual remote residues.
Block summation, Theorem 3.4	Finite block geometry, canceled regular factors, divided differences, contour exhaustion	Finite d and positive real periods.
Raw abscissa, Theorem 4.2	Explicit residues, root test, reciprocal approximation identity	Irrational two-lattice ratio; boundary line unclassified.
Finite-cut uniqueness, Theorem 5.1	Block tail and finite splitting of the cutoff block	Asymptotic statement; not raw-series convergence.
Coalescence, Theorem 6.1	Affine divided-difference scaling and the simplex integral	Compact scaled offsets and bounded regular-factor derivatives.
Compression, Theorem 7.2	Periodic polynomial sequences and action uniqueness	Necessity concerns finite single-base rational forms in $d = 2$.
Inverse certificate, Theorem 8.1	Classical contour Lagrange inversion applied in the block norm	Local certified branch, not global inverse univalence.
Inverse action formula, Theorem 8.3	Positive-action filtration, finite coefficient extraction, analytic tails	No complete inverse Borel continuation theorem.
Abstract criterion, Proposition 9.1	Local canceled-factor bounds and explicit contour hypotheses	Does not manufacture those hypotheses for a general kernel.

All theorem statements in this report have conventional proofs. The computations corroborate selected identities and examples but are not a formal proof assistant or a numerical substitute for the infinite estimates. In particular, the report does not claim to settle a named community-wide conjecture, establish priority over all special-function literature, or supply a Lean-checked development.

B Source provenance and reproduction

The repository was inspected through its GitHub connection. The frozen snapshot used for the explicit comparison is

0c973d8f5e7ef4550f4df1bf486d890037fd9f14.

The most directly relevant file is

Analysis/Transseries/docs/series-and-transseries/
Nonlinear_Stokes_Transport_Logarithmic_Inversion/article.tex

The inspected source lines 180–360 include the contour proof, derivative-disk certificate, weighted countable extension, collision proposition, and exponential-action transport discussion. The neighboring README states that its countable analytic extension is not by itself a countable-action resurgence theorem. The comparison in Section 1 refers to these specific statements. The inventory was also consulted to avoid repeating the recent moving-fold, countable-feedback, harmonic-inverse, and action-accumulation topics. Repository documents were treated as mathematical sources to inspect, not as automatically verified facts.

The package has no dependency on a local checkout of ProveIt. The bibliography is embedded in this source, and the one numerical table is included as a generated TeX file. To reproduce the checks and the PDF:

```
python -m pip install -r requirements.txt
python verify.py --stage all
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The supplied `build.sh` performs the same steps after changing to its own directory. The verification program accesses no network services and writes only local output files. See `SOURCES.md` for direct source links and `QA_REPORT.md` for the actual PDF build and visual inspection record. No external source corpus or font files are included.

References

- [1] Vladimir Reshetnikov’s ProveIt repository, *Nonlinear Stokes Transport under Logarithmic Inversion*, research article and README, 29 September 2026. Snapshot 0c973d8f5e7ef4550f4df1bf486d890037fd9f14; source path recorded in Appendix B. <https://github.com/VladimirReshetnikov/ProveIt>.
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